

# Spectral Likelihood Ratios for Nearly Optimal Community Recovery in the Stochastic Block Model

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October 9, 2026  
Research manuscript

## Abstract

Spectral clustering compresses a network into a small number of eigenpairs, whereas likelihood-ratio classification uses block-dependent logarithmic contrasts. We study how to decode the latter information from the former representation. For a stochastic block model with a full-rank connectivity matrix, the population signal space contains every block-constant likelihood contrast. An exact projection identity explains why spectral compression can preserve the linear part of an oracle likelihood ratio. We propose a spectral-only procedure that selects an initial profile on a high-likelihood subset, adds profiles according to their joint gain on all nodes, and fits a positive-profile working mixture by alternating updates. We prove a conditional recovery theorem with error bound  $e^{-(1+o(1))I_n}$ , together with exact contrast-preservation, sparse likelihood approximation, greedy-selection, and objective-convergence results. The recovery theorem requires likelihood-score errors and their exceptional-event probabilities to be controlled at the oracle error scale; establishing these conditions for the complete empirical procedure remains unresolved. On two paired collections totaling 240 simulated graphs, global-gain seeding or likelihood-accepted parameter replacement eliminates two severe initialization failures and reduces mean misclassification relative to the tested spectral  $k$ -means baseline. We also formulate the lower-density extension and identify the additional calibration required by regularized spectral representations.

## 1 Introduction

Community recovery in the stochastic block model (SBM) is a basic problem in network statistics. The model assigns each node to an unknown community and makes edge probabilities depend on the communities of their endpoints [6]. Two common approaches use different representations of the same data. Spectral methods reduce the adjacency matrix to a small number of eigenvectors and then cluster the embedded nodes. Likelihood methods compare how well alternative community assignments explain the observed connections. The question studied here is whether the likelihood-ratio information needed for accurate classification remains available after spectral reduction, without returning to the original adjacency rows.

This question concerns the decoding rule as well as the embedding. A low-rank approximation is selected by a quadratic criterion, and standard spectral clustering applies a further Euclidean criterion through  $K$ -means. The oracle classifier for an SBM instead weights connections by logarithms of block probabilities and includes the contribution of absent edges. For a general connectivity matrix and unequal community sizes, its decision boundaries need not agree with nearest-center boundaries in the spectral embedding. Nevertheless, the fact that spectral reduction

uses a quadratic objective does not itself imply that it discards the relevant likelihood information: a projection preserves every linear statistic whose weight vector belongs to the projected subspace.

**Optimal recovery and likelihood refinement.** The information-theoretic limits of the SBM are well developed. Abbe and Sandon [1] characterize exact recovery in the general logarithmic-density SBM through Chernoff–Hellinger divergence and give efficient algorithms reaching that threshold. Zhou and Li [12] study the optimal error rate through Chernoff bounds and its application to community detection. We use these works to identify the oracle benchmark, while the statistical objective of the present paper is the leading-exponent form

$$\exp\{-(1 + o(1))I\},$$

where  $I$  is the relevant pairwise Chernoff information. We do not claim the more precise optimal rate investigated by Zhou and Li.

Spectral initialization followed by likelihood refinement is also an established approach. Amini, Chen, Bickel and Levina [3] fit mixtures to blockwise connection counts using pseudo-likelihood and iterative updates. Gao, Ma, Zhang and Zhou [5] prove that a weakly consistent initializer followed by penalized local likelihood refinement achieves optimal misclassification performance under their assumptions. Zhou and Amini [11] obtain optimal bipartite recovery using spectral initialization and pseudo-likelihood classification, and analyze a broader family of EM-type biclustering updates. In these methods the refinement uses information calculated from the original edge observations. Our algorithm imposes an additional restriction: once the selected adjacency eigenpairs have been computed, initialization, parameter updates, classification and component replacement use only those eigenpairs and quantities reconstructed from them.

**Likelihood information in spectral representations.** Spectral optimality is known in important special cases. Abbe, Fan, Wang and Zhong [2] establish a first-order entrywise approximation for eigenvectors and show that, in the equal-sized symmetric two-community SBM at logarithmic density, an untrimmed spectral method attains the optimal recovery threshold and misclassification exponent. Their result already demonstrates that a spectral representation can retain the decisive local evidence. For sparse homogeneous SBMs with multiple, possibly unequal communities, Zhang [9] gives matching upper and lower error exponents for adjacency spectral clustering with  $K$ -means. His spectral exponent is distinguished from the information-theoretic exponent, so a sharp analysis of a spectral algorithm does not automatically establish its statistical optimality.

Probabilistic inference based on spectral embeddings predates the present work. Suwan et al. [8] use Gaussian-mixture limits of adjacency spectral embedding to construct an empirical Bayes prior for block membership inference. This is a relevant example of using distributional information beyond Euclidean clustering; it does not establish a sparse large-deviation equivalence between our spectral scores and the oracle edge likelihood ratio. For lower densities, Zhou and Amini [10] provide data-driven adjacency regularization and consistency analysis in the general bipartite setting. These results supply useful tools for the extension considered in Section 4, but regularized spectral concentration by itself does not prove preservation of the oracle error exponent.

**Our approach.** We retain the  $K$  eigenpairs of the adjacency matrix with largest absolute eigenvalues, including signal eigenvalues of either sign, and form reconstructed connection profiles. A Poisson-form working score compares each profile with candidate community profiles. Its pairwise difference has the algebraic form of a likelihood ratio. We initialize the first component by its score on

a restricted subset of nodes, then choose further components by the increase in the best score over all nodes. The latter criterion gives little reward for repeatedly representing an already well-explained part of the data. We also study a component-replacement step that accepts a proposed replacement only after reoptimization improves the common working objective. These choices are motivated by observed failures in which two nearby communities are represented by one intermediate profile while another community receives two components.

The interpretation as a spectral likelihood ratio has a precise algebraic basis. If  $\hat{\Pi}$  denotes the projector onto the retained empirical eigenspace, its signed rank- $K$  reconstruction equals  $A\hat{\Pi}$ . With a population SBM projector, every block-constant likelihood-ratio contrast is preserved when the block matrix is full rank. Empirical projection, positivity clipping and estimated parameters introduce additional errors. The reconstructed profiles are dependent, noninteger quantities, so the working objective is not the probability likelihood of independent Poisson observations. We therefore distinguish the numerical fitting objective from the oracle Bernoulli likelihood and state the statistical approximation required to connect them.

The paper makes three contributions. First, it gives an explicit spectral-only fitting algorithm together with algebraic identities and optimization guarantees: a diminishing-returns property for frozen-candidate greedy selection and monotonicity of exact EM updates. Second, it proves a conditional recovery theorem: sufficiently accurate pairwise approximation to the oracle likelihood ratios, with exceptional-event probabilities controlled at the relevant error scale, transfers the leading Chernoff exponent. This theorem identifies exactly what must be proved for an unconditional guarantee for the executable algorithm; the complete clipped-profile EM pipeline is not asserted to satisfy that condition here. Third, experiments compare initialization and refinement variants on logarithmic-density SBMs, diagnose severe local optima, and test whether global score gains or component replacement resolve them without using true labels during fitting.

Section 2 specifies the model, spectral scores and algorithms. Section 3 gives the main results without adjacency regularization in the logarithmic-density regime. Section 4 formulates the lower-density extension and its additional requirements. Section 5 presents experiments and failure diagnostics, and Section 6 concludes. Proofs appear in the appendix.

## 2 Spectral likelihood-ratio decoding

### 2.1 Model and retained information

Let  $A$  be the adjacency matrix of an undirected graph without self-loops. Conditional on community labels  $z = (z_1, \dots, z_n) \in [K]^n$ , its upper-triangular entries are independent and

$$A_{ij} \sim \text{Bernoulli}(\rho_n B_{z_i z_j}), \quad i < j, \quad A_{ji} = A_{ij}, \quad A_{ii} = 0. \quad (2.1)$$

The number  $K$  is fixed and known. Write  $Z_{ia} = \mathbf{1}\{z_i = a\}$ ,  $n_a = \sum_i Z_{ia}$ , and  $\pi_{a,n} = n_a/n$ . Unless specified otherwise,  $B$  is a fixed symmetric full-rank matrix,  $0 < b_- \leq B_{ab} \leq b_+ < \infty$ , and  $\pi_{a,n} \rightarrow \pi_a > 0$ . We begin with  $\rho_n = \log n/n$  and put  $d_n = n\rho_n$ . Full rank need not imply positive definiteness; disassortative signal directions are retained.

The actual mean matrix and its rank- $K$  comparison are

$$P = \mathbb{E}[A \mid z] = P^\circ - \text{diag}(\rho_n B_{z_i z_i}), \quad P^\circ = \rho_n Z B Z^\top. \quad (2.2)$$

The small diagonal correction matters for exact identities. The representation consists of the  $K$

empirical eigenpairs of  $A$  with largest absolute eigenvalues:

$$AU = U\Lambda, \quad U^\top U = I_K, \quad \tilde{P} = U\Lambda U^\top. \quad (2.3)$$

All subsequent fitting accesses only  $(U, \Lambda)$ . This restriction excludes an additional edge-based refinement. It also excludes computing separate leave-one-out eigendecompositions as an undeclared algorithmic step. In the logarithmic regime we use the adjacency eigenpairs without degree regularization or separately stored degrees.

## 2.2 Positive profiles and a working likelihood

We construct the positive profiles

$$q_{ij} = \max\{\tilde{P}_{ij}, \varepsilon_n\}, \quad \varepsilon_n = 10^{-4} \log n/n. \quad (2.4)$$

The evaluated implementation retains the reconstructed diagonal. For  $\mu \in (0, \infty)^n$  define

$$\ell_i(\mu) = \sum_{j=1}^n q_{ij} \log \mu_j - \sum_{j=1}^n \mu_j. \quad (2.5)$$

Terms independent of  $\mu$  have been omitted. This convention fixes the first-seed subset scores: restoring observation-only Gamma constants can change which subset wins, even though it does not change mixture fitting or subsequent global gains. Although this is the algebraic form of a Poisson log likelihood,  $q_{ij}$  are dependent, generally noninteger quantities. We therefore use it as a *working* score, rather than assert that the spectral observations follow an independent Poisson model.

For profiles  $\mu_1, \dots, \mu_K$  and weights  $\omega \in \Delta_K$ , the working mixture objective is

$$\mathcal{L}(\omega, \mu) = \sum_{i=1}^n \log \left( \sum_{a=1}^K \omega_a e^{\ell_i(\mu_a)} \right). \quad (2.6)$$

The final label is

$$\hat{z}_i = \arg \max_{a \in [K]} \{\ell_i(\mu_a) + \log \omega_a\}. \quad (2.7)$$

Thus classification uses pairwise working likelihood ratios, with estimated prior weights determining the thresholds. We evaluate the permutation-invariant mislabeled fraction

$$r(\hat{z}, z) = \min_{\sigma \in \mathfrak{S}_K} \frac{1}{n} \sum_{i=1}^n \mathbf{1}\{\sigma(\hat{z}_i) \neq z_i\}, \quad (2.8)$$

where  $\mathfrak{S}_K$  is the set of label permutations. The unrestricted profiles are  $n$ -dimensional weighted means; the implementation does not impose the exact SBM constraint  $\mu_{aj} = \rho_n B_{a,z_j}$ .

## 2.3 Subset-first, global-gain initialization

Every observed profile  $q_c$  is a candidate parameter. The first seed uses a restricted-subset score. Let

$$m = \min \left\{ \left\lfloor \frac{n}{1.5K} \right\rfloor, n - K \right\}, \quad T_c \subset [n] \setminus \{c\}, \quad |T_c| = m, \quad (2.9)$$

where  $T_c$  contains the nodes having largest values of  $\ell_i(q_c)$ . Choose

$$c_1 \in \arg \max_c \sum_{i \in T_c} \ell_i(q_c), \quad \mu_1 = q_{c_1}. \quad (2.10)$$

This step looks for a candidate that fits a substantial subset rather than compromises across the whole graph. The candidate itself is excluded from its evaluation subset. Ties are broken deterministically.

Subsequent steps keep every node and assess how much a new profile improves their joint explanation. With  $h_i^{(t-1)} = \max_{a < t} \ell_i(\mu_a)$ , choose

$$c_t \in \arg \max_c \Delta(c \mid \mu_{1:t-1}), \quad \Delta(c \mid \mu_{1:t-1}) = \sum_i (\ell_i(q_c) - h_i^{(t-1)})_+, \quad \mu_t = q_{c_t}. \quad (2.11)$$

No subset-size constraint is imposed after the first seed, and no nodes are removed. Selecting a profile already represented in the current set contributes zero gain. A mixed profile can still have positive gain, so this rule does not guarantee that every seed is a community center. Its purpose is to make repeated coverage compete against the improvement available in omitted regions.

This is a deterministic likelihood-gain analogue of the residual-coverage principle behind  $k$ -means++ [4]. Unlike  $D^2$  sampling, it selects the largest gain rather than sampling proportionally to a distance. Proposition B.1 proves a standard submodular guarantee relative to the fixed first seed, and Proposition B.2 proves complete coverage in the noiseless positive-profile model.

## 2.4 Alternating fitting and parameter replacement

After  $K$  seeds have been selected, initialize  $\omega_a = 1/K$  and perform soft updates

$$\tau_{ia} = \frac{\omega_a e^{\ell_i(\mu_a)}}{\sum_b \omega_b e^{\ell_i(\mu_b)}}, \quad N_a = \sum_i \tau_{ia}, \quad (2.12)$$

$$\mu_{aj}^{\text{new}} = \frac{\sum_i \tau_{ia} q_{ij}}{N_a}, \quad \omega_a^{\text{new}} = N_a/n. \quad (2.13)$$

The responsibilities are computed using log-sum-exp for numerical stability. Positive  $q$  ensures that positive-weight components have positive updated means. The evaluated implementation constrains  $\omega_a \geq 10^{-8}$  and replaces the unconstrained weight update in (2.13) by the exact optimizer of  $\sum_a N_a \log \omega_a$  on that floored simplex. If no floor is active, this reduces to  $N_a/n$ . Both exact versions maximize the variational M-step, giving objective monotonicity and boundedness. Proposition B.3 proves objective convergence; it does not prove global optimality or correct community recovery.

### Algorithm 1: Spectral-only likelihood-gain decoding

1. Input  $K$  and the retained eigenpairs  $(U, \Lambda)$ ; form  $q$  by (2.4).
2. Select the first profile by (2.10), with  $m = \lfloor n/(1.5K) \rfloor$ .
3. For  $t = 2, \dots, K$ , add the candidate maximizing (2.11) on all  $n$  nodes, keeping the previously selected profiles fixed.
4. Run the joint updates (2.12)–(2.13) to the declared objective tolerance or iteration limit.
5. Output the labels in (2.7), the fitted profiles, weights, and objective trace.

We also evaluate an alternative stabilization of the earlier partial-peeling initialization. After fitting, remove each component in turn; choose a replacement profile by its global gain relative to

the remaining fitted profiles, and run a fresh fit. Accept the new fit only if its final working mixture objective increases. The evaluated replacement pass tests the  $K$  possible component removals and selects an improving trial. This is an alternative branch with additional fitting cost; the results below do not assert that the combination of Algorithm 1 and replacement was evaluated.

The numerical experiments use the exact floored-simplex weight update, finite stopping tolerances, and an iteration cap. Objective convergence in exact arithmetic does not replace checking the finite-run stopping rule or floating-point error. We report those numerical checks separately.

## 2.5 Why the representation can retain an LLR

Suppose temporarily that  $B$  and the other labels are known. For testing  $z_i = a$  against  $z_i = b$ , define, for  $j \neq i$ ,

$$w_j^{ab} = \log \frac{B_{a,z_j}(1 - \rho_n B_{b,z_j})}{B_{b,z_j}(1 - \rho_n B_{a,z_j})}, \quad c_i^{ab} = \sum_{j \neq i} \log \frac{1 - \rho_n B_{a,z_j}}{1 - \rho_n B_{b,z_j}}. \quad (2.14)$$

The oracle Bernoulli LLR is  $L_i^{ab} = \sum_{j \neq i} A_{ij} w_j^{ab} + c_i^{ab}$ . Extend  $w_i^{ab}$  by the same block formula. Because  $A_{ii} = 0$ , this extension does not change the oracle linear statistic. The contrast vector is block constant, hence belongs to  $\text{col}(Z)$ .

The fundamental identity is

$$\tilde{P} = A\Pi, \quad \Pi = UU^\top, \quad (\tilde{P}_i - A_i)w^{ab} = A_i(\Pi - I)w^{ab}. \quad (2.15)$$

Here  $I$  denotes the identity matrix. If  $w^{ab}$  lies in the retained empirical space, the linear statistic is preserved exactly. At the population level, full rank gives  $\text{col}(P^\circ) = \text{col}(Z)$ , so every contrast has the required span membership. The spectral decomposition need not itself optimize a likelihood: it retains a linear space containing the directions in which the likelihood acts.

The executable score additionally involves clipping, estimated profiles, mixture thresholds, and a retained reconstructed diagonal. These perturbations require statistical control. In particular, neither low-rank reconstruction error nor objective convergence alone establishes that (2.7) achieves the oracle error exponent. Section 3 states an explicit sufficient theorem for that conclusion.

## 2.6 Computational cost

Forming the full positive reconstruction stores  $O(n^2)$  numbers. Caching all candidate scores uses an  $n$ -by- $n$  matrix and, by direct multiplication,  $O(n^3)$  arithmetic. Each subsequent greedy scan costs  $O(n^2)$ , and one EM iteration costs  $O(Kn^2)$ . These costs are additional to computing the eigenpairs. The implementation is practical at the tested  $n \leq 2000$  but does not yet justify a large-network speed claim. Candidate subsampling and blocked score evaluation are natural approximations, whose statistical and computational performance require separate validation.

## 3 Main results in logarithmic density

This section separates an exact information-preservation identity, a statistical exponent-transfer theorem, and a sufficient condition for an independent-row spectral projection. The exponent-transfer theorem applies to the fitted working scores in Section 2 *if its stated approximation condition holds*. We do not assert that convergence of the working EM objective establishes that condition.

Throughout this section,  $K$  is fixed,  $n_a/n \rightarrow \pi_a > 0$ ,  $B$  is a fixed symmetric matrix with  $0 < b_- \leq B_{ab} \leq b_+ < \infty$ , and  $\rho_n \rightarrow 0$  with  $d_n = n\rho_n \rightarrow \infty$ . Distinct rows of  $B$  are assumed whenever classification between all communities is discussed. The logarithmic regime is  $\rho_n = \log n/n$ . Full rank of  $B$  is used for the signal-span interpretation; the probabilistic oracle calculations do not require it.

### 3.1 Spectral information and likelihood contrasts

For communities  $a \neq b$ , define the block-constant Bernoulli contrast and the nonedge offset by

$$w_j^{ab} = \log \frac{B_{a,z_j}(1 - \rho_n B_{b,z_j})}{B_{b,z_j}(1 - \rho_n B_{a,z_j})}, \quad (3.1)$$

$$c_i^{ab} = \sum_{j \neq i} \log \frac{1 - \rho_n B_{a,z_j}}{1 - \rho_n B_{b,z_j}}. \quad (3.2)$$

The oracle incident-edge log-likelihood ratio is

$$L_i^{ab} = A_i w^{ab} + c_i^{ab}, \quad (3.3)$$

where  $A_i$  is row  $i$  of  $A$ . Its diagonal coordinate contributes zero.

**Proposition 3.1** (Exact preservation of retained contrasts). *Let  $U$  contain any orthonormal collection of eigenvectors of the symmetric matrix  $A$ , let  $\Lambda$  contain their eigenvalues, and set  $Q = UU^\top$  and  $\tilde{P} = U\Lambda U^\top$ . Then  $\tilde{P} = AQ$  and, for every vector  $w$ ,*

$$\tilde{P}_i w - A_i w = A_i(Q - I)w. \quad (3.4)$$

*In particular,  $w \in \text{col}(U)$  implies exact preservation. Moreover, if  $B$  is nonsingular, then the rank- $K$  comparison matrix  $P^\circ = \rho_n Z B Z^\top$  has column space  $\text{col}(Z)$ , and every  $w^{ab}$  belongs to this population signal space.*

The proposition explains the likelihood interpretation without requiring the eigendecomposition itself to compute  $\log B$ . The eigendecomposition retains directions; the likelihood decoder assigns the appropriate contrast coefficients to those directions. It also states the precise limitation: the empirical retained space need not equal the population space. The actual no-self-loop mean is  $P = P^\circ - \text{diag}(\rho_n B_{z_i z_i})$ ; the rank- $K$  statement is about  $P^\circ$ , not  $P$ .

The oracle Poisson contrast is  $v_j^{ab} = \log(B_{a,z_j}/B_{b,z_j})$ , with score

$$L_{i,P}^{ab} = \sum_{j \neq i} A_{ij} v_j^{ab} - \rho_n \sum_{j \neq i} (B_{a,z_j} - B_{b,z_j}). \quad (3.5)$$

The following result makes its use in a sparse working score precise.

**Lemma 3.2** (Sparse Bernoulli-to-Poisson correction). *There is a constant  $C$  depending only on  $b_-, b_+$  such that, for all sufficiently large  $n$ , every  $i$ , and every  $a \neq b$ ,*

$$|L_i^{ab} - L_{i,P}^{ab}| \leq C\{\rho_n \deg(i) + n\rho_n^2\}. \quad (3.6)$$

*For every fixed  $H > 0$  there is a constant  $D_H$  such that  $\Pr\{\deg(i) > D_H d_n\} \leq e^{-H d_n}$  uniformly in  $i$ . Consequently, the correction is  $o(d_n)$  outside an event of probability  $e^{-H d_n}$ . In logarithmic density its bound on that event is  $O(\log^2 n/n) = o(1)$ .*

### 3.2 The oracle benchmark

Let

$$J_{ab} = \max_{0 \leq t \leq 1} \sum_{c=1}^K \pi_c \{ (1-t)B_{ac} + tB_{bc} - B_{ac}^{1-t} B_{bc}^t \}, \quad (3.7)$$

$$J_* = \min_{a \neq b} J_{ab}, \quad I_n = d_n J_*. \quad (3.8)$$

Equivalently,  $J_{ab}$  is the Chernoff–Hellinger divergence between  $(\pi_c B_{ac})_c$  and  $(\pi_c B_{bc})_c$ , with an interchangeable choice of  $t$  and  $1-t$ . Our recovery target concerns the leading error exponent. Zhou and Li [12] studied optimal error rates for SBM community detection; their more refined rate expressions are not needed for the claims made here.

**Lemma 3.3** (Leading oracle error exponent). *Consider the binary experiment in which  $z_i \in \{a, b\}$  has equal prior probabilities and all other labels and  $B$  are known. Its Bayes risk  $R_{i,ab}^{\text{or}}$  satisfies, uniformly in  $i$ ,*

$$R_{i,ab}^{\text{or}} = \exp\{-(1+o(1))d_n J_{ab}\}. \quad (3.9)$$

The same leading exponent holds for fixed prior probabilities bounded away from zero and one.

### 3.3 Transfer to estimated spectral scores

The theorem below accommodates scores estimated from the entire graph, so it does not assume independence of a node's score and its incident edges. It makes the required rare-event control explicit.

**Theorem 3.4** (Conditional nearly-optimal exponent). *Let  $\hat{\ell}_i(a)$  be estimated community scores, after one common alignment of their labels with the population labels, and classify by  $\hat{z}_i \in \arg \max_a \hat{\ell}_i(a)$ . Suppose there are  $\tau_n = o(d_n)$  and  $\delta_n \geq 0$  such that, for every  $i$ , an event  $G_i$  satisfies*

$$\Pr(G_i^c) \leq \delta_n, \quad (3.10)$$

$$\max_{a \neq b} |\hat{\ell}_i(a) - \hat{\ell}_i(b) - L_i^{ab}| \leq \tau_n \quad \text{on } G_i, \quad (3.11)$$

$$\liminf_{n \rightarrow \infty} \frac{-\log \delta_n}{I_n} \geq 1, \quad (3.12)$$

where  $-\log 0 = +\infty$ . Then the permutation-invariant error fraction obeys

$$\mathbb{E} r(\hat{z}, z) \leq \exp\{-(1+o(1))I_n\}. \quad (3.13)$$

When  $\rho_n = \log n/n$  and  $J_* > 1$ , this implies  $\Pr\{r(\hat{z}, z) > 0\} \rightarrow 0$ .

The condition on  $\delta_n$  is essential. An error approximation that holds with probability  $1 - o(1)$  can still fail much more frequently than the oracle makes an error, and therefore need not preserve its exponent. The theorem asserts an upper bound on the recovery error. In the binary oracle experiment, data processing and Lemma 3.3 also give the matching leading lower bound for any decoder using only the graph's spectral summary.

For the working scores actually fitted in Section 2, set  $\hat{w}_j^{ab} = \log(\hat{\mu}_{aj}/\hat{\mu}_{bj})$  and  $\hat{c}^{ab} = -\sum_j (\hat{\mu}_{aj} - \hat{\mu}_{bj}) + \log(\hat{\omega}_a/\hat{\omega}_b)$ . Writing  $q_i = \tilde{P}_i + (q_i - \tilde{P}_i)$  yields the exact decomposition

$$\begin{aligned} \hat{\ell}_i(a) - \hat{\ell}_i(b) - L_i^{ab} &= A_i(Q - I)w^{ab} \\ &\quad + \tilde{P}_i(\hat{w}^{ab} - w^{ab}) \\ &\quad + (q_i - \tilde{P}_i)\hat{w}^{ab} + (\hat{c}^{ab} - c_i^{ab}). \end{aligned} \quad (3.14)$$

Thus projection, estimated contrasts, clipping, and offsets must all be controlled at the scale in Theorem 3.4. The sum includes the reconstructed diagonal used by the implemented working score; its discrepancy with the oracle's excluded diagonal is included in the displayed decomposition. Neither average reconstruction accuracy nor objective convergence removes these terms.

### 3.4 A sufficient independent-row projection condition

The next theorem proves a usable bridge between subspace estimation and rare-event LLR accuracy. It is a result about a row-independent projection, not an unconditional guarantee for the full-graph fitted algorithm. Leave-one-out projections provide a natural setting for this analysis.

**Theorem 3.5** (Row-independent spectral projection). *Fix  $i$  and restrict vectors and projectors to coordinates  $j \neq i$ . Let  $Q_i$  be a random orthogonal projector independent of  $(A_{ij})_{j \neq i}$ , and let  $Q_{*,i}$  project onto the membership space of the other nodes. Suppose that on an event  $H_i$ , measurable with respect to the data generating  $Q_i$ ,*

$$\|(I - Q_i)Q_{*,i}\|_{\text{op}} \leq \eta_n \rightarrow 0, \quad \max_{j \neq i} (Q_i)_{jj} \leq \frac{L}{n}, \quad (3.15)$$

for a constant  $L$ , and that  $\Pr(H_i^c) \leq \gamma_n$ . Define

$$L_{i,Q}^{ab} = A_{i,-i}Q_i w_{-i}^{ab} + c_i^{ab}. \quad (3.16)$$

There are deterministic  $\tau_n = o(d_n)$  and  $h_n \rightarrow \infty$  such that

$$\Pr \left\{ \max_{a \neq b} |L_{i,Q}^{ab} - L_i^{ab}| > \tau_n \right\} \leq \gamma_n + 2K^2 e^{-h_n d_n}. \quad (3.17)$$

Consequently, if  $\liminf(-\log \gamma_n)/I_n \geq 1$  uniformly in  $i$ , classifiers based on these oracle-parameter projected scores satisfy (3.13).

The proof uses a quadratic population bias and a Bennett tail bound. It requires bounded projector leverage, rather than a vanishing maximum coordinate error. Its independence assumption is explicit: forming  $A_{i,-i}Q_i$  directly is not an additional step of our post-compression algorithm. To apply the theorem to that algorithm, one must further control the difference between its empirical full-graph projection and an independent-row construction, and then the remaining terms in (3.14). Those pipeline conditions are not established by the present initialization and EM results.

## 4 Extension below logarithmic density

### 4.1 Recovery target

Consider  $d_n = n\rho_n \rightarrow \infty$  with  $d_n = o(\log n)$ , keeping  $K$ , positive limiting community proportions, and  $B$  fixed. The pairwise population profiles remain distinct, and the leading oracle information is proportional to  $d_n$ . The appropriate target is a vanishing fraction of errors with bound

$$\mathbb{E}r(\hat{z}, z) \leq e^{-(1+o(1))I_n}, \quad I_n = d_n J_*. \quad (4.1)$$

For consistency the right-hand side tends to zero. It generally does not yield exact recovery when  $d_n = o(\log n)$ . The distinction concerns the recovery objective, rather than a change in the meaning of the likelihood-ratio benchmark.

The score-transfer theorem in Section 3 applies at this density whenever its score-error conditions hold. The sparse Bernoulli-to-Poisson approximation remains valid because its exponent-scale correction is  $O(n\rho_n^2) = o(d_n)$ . These facts provide a precise route for extension. They do not by themselves control the empirical eigenspace.

## 4.2 Representation and calibration

Low-degree concentration is more delicate: a few atypically high-degree vertices can produce localized empirical directions. Regularization can stabilize a spectral initializer, but a regularized representation must also retain the directions used for likelihood classification. Existing regularized spectral consistency results [10] concern recovery or concentration under their stated conditions; they do not automatically imply exponentially accurate spectral likelihood decoding.

For example, suppose a positive diagonal normalization  $D$  is used and the retained eigenpairs satisfy

$$M = D^{-1/2}AD^{-1/2}, \quad MV = V\Gamma, \quad \Pi_M = VV^\top. \quad (4.2)$$

The reconstruction in the original scale is

$$\tilde{A}_D = D^{1/2}V\Gamma V^\top D^{1/2} = AD^{-1/2}\Pi_M D^{1/2}. \quad (4.3)$$

Consequently  $\tilde{A}_D w = Aw$  whenever  $D^{1/2}w \in \text{col}(V)$ . This exact identity exhibits the needed transformed contrast. The normalization factors must be retained with the eigenpairs to perform this reconstruction. A normalized embedding without its scale factors is not covered by this identity.

Trimming is a different operation. If  $M = A - T$  is obtained by modifying entries, its retained reconstruction is  $M\Pi_M$ , and

$$A_i w - (M\Pi_M)_i w = T_i w + M_i(I - \Pi_M)w. \quad (4.4)$$

Both the removed-edge contrast and projection error must be controlled at the recovery scale. This prevents interpreting every regularized spectral embedding as automatically equivalent to the original-graph LRT.

## 4.3 A conditional extension

**Corollary 4.1** (Lower-density transfer). *Under the fixed- $B$  model with  $d_n \rightarrow \infty$ ,  $\rho_n \rightarrow 0$ , and positive limiting proportions, let a spectral decoder, possibly calibrated as in (4.3), satisfy the score approximation and exceptional-event bounds of Theorem 3.4. Then its expected misclassification rate satisfies (4.1). If  $d_n = o(\log n)$ , this statement establishes almost exact recovery, without an exact-recovery conclusion.*

*Proof.* The Chernoff expansion and score-transfer proof use  $d_n \rightarrow \infty$  and  $n\rho_n^2 = o(d_n)$ , which follow here from  $\rho_n \rightarrow 0$ . Substitute the density into Theorem 3.4. Since the minimum divergence is strictly positive for distinct profiles,  $I_n \rightarrow \infty$  and the expected fraction of errors tends to zero.  $\square$

Verifying this corollary for a specified regularizer and fitting procedure remains a separate statistical task. The present 240-graph evaluation uses the unregularized logarithmic-density representation. Bounded expected degree, where detection has a different objective and threshold, is outside the scope of these results.

## 5 Experiments

We evaluate whether likelihood-based decoding improves on a Euclidean decoder of the same retained eigenpairs, and whether global likelihood gains prevent redundant initialization. The experiments use a spectral profile working likelihood. They do not evaluate an exact leave-one-out graph likelihood or estimate an asymptotic error exponent from finite samples.

### 5.1 Models and protocol

For each model, let  $C^{(0)}$  be a fixed symmetric connectivity matrix and let  $\pi$  be the vector of community proportions. We generate an undirected SBM without self-loops, with edge probabilities

$$(B_n)_{ab} = \frac{\log n}{n} C_{ab}, \quad C = \frac{\eta}{D_{\min}(C^{(0)}, \pi)} C^{(0)}, \quad (5.1)$$

where  $\eta \in \{0.8, 1.4\}$  and

$$D_{\min}(C, \pi) = \min_{a \neq b} \max_{0 \leq t \leq 1} \sum_{c=1}^K \pi_c \left[ (1-t)C_{ac} + tC_{bc} - C_{ac}^{1-t} C_{bc}^t \right]. \quad (5.2)$$

Thus the limiting minimum Chernoff–Hellinger coefficient equals  $\eta$ . The normalization uses the limiting proportions  $\pi$ ; actual group sizes are  $\lfloor n\pi_a \rfloor$ , with the remaining vertices assigned to the last group. Vertex labels are then randomly permuted. Each model is tested at  $n \in \{1000, 2000\}$ . The six models are

$$\begin{aligned} C_{\text{sym2}}^{(0)} &= \begin{pmatrix} 4 & 1 \\ 1 & 4 \end{pmatrix}, & \pi_{\text{sym2}} &= (1/2, 1/2), \\ C_{\text{equal3}}^{(0)} &= \begin{pmatrix} 10 & 1 & 2 \\ 1 & 11 & 1 \\ 2 & 1 & 10 \end{pmatrix}, & \pi_{\text{equal3}} &= (1/3, 1/3, 1/3), \\ C_{\text{hetero3}}^{(0)} &= \begin{pmatrix} 6 & 0.5 & 0.5 \\ 0.5 & 12 & 0.5 \\ 0.5 & 0.5 & 24 \end{pmatrix}, & \pi_{\text{hetero3}} &= (1/3, 1/3, 1/3), \\ C_{\text{unequal3}}^{(0)} &= \begin{pmatrix} 10 & 1 & 1 \\ 1 & 10 & 1 \\ 1 & 1 & 10 \end{pmatrix}, & \pi_{\text{unequal3}} &= (0.5, 0.3, 0.2), \\ C_{\text{hier4}}^{(0)} &= \begin{pmatrix} 12 & 2 & 0.5 & 0.5 \\ 2 & 12 & 0.5 & 0.5 \\ 0.5 & 0.5 & 12 & 2 \\ 0.5 & 0.5 & 2 & 12 \end{pmatrix}, & \pi_{\text{hier4}} &= (1/4, 1/4, 1/4, 1/4), \\ C_{\text{dis3}}^{(0)} &= \begin{pmatrix} 0.5 & 8 & 8 \\ 8 & 0.5 & 8 \\ 8 & 8 & 0.5 \end{pmatrix}, & \pi_{\text{dis3}} &= (1/3, 1/3, 1/3). \end{aligned}$$

These include unequal group sizes, heterogeneous expected degrees, hierarchical connections, and negative signal eigenvalues.

The design batch contains five graph seeds, 20000,  $\dots$ , 20004, for every model, size, and signal coefficient, giving  $6 \times 2 \times 2 \times 5 = 120$  graphs. A second batch uses the same models and hyperparameters with fresh seeds 30000,  $\dots$ , 30004. The algorithms and second-batch seed range were fixed before fitting the second batch. The second batch therefore validates initialization choices made using the first batch, rather than introducing new models. All methods are evaluated on identical eigenpairs within a graph.

We retain the  $K$  eigenpairs of the raw adjacency matrix with largest absolute eigenvalues. The implementation requests  $K + 1$  eigenpairs using the symmetric ARPACK solver with tolerance  $10^{-8}$  and discards the last one after sorting by absolute value. Neither spectral regularization nor degree rescaling is used. After this preprocessing, the fitting routines receive only  $(U, \Lambda)$ ; adjacency entries, degrees, true labels, and true block probabilities are unavailable to fitting and model selection.

## 5.2 Implementations and comparisons

The archived implementation forms positive spectral profiles

$$q_{ij} = \max \left\{ (U \Lambda U^\top)_{ij}, 10^{-4} \frac{\log n}{n} \right\}, \quad \ell_i(\mu) = \sum_{j=1}^n q_{ij} \log \mu_j - \sum_{j=1}^n \mu_j. \quad (5.3)$$

The reconstructed diagonal is retained, even though the generated graphs have zero diagonal. Thus this is a declared positive-profile working score, not the original Bernoulli graph likelihood. Component means are unrestricted  $n$ -dimensional profile means. Every likelihood arm uses the same soft EM updates and observed working objective

$$\mathcal{L}(\mu, w) = \sum_{i=1}^n \log \left[ \sum_{a=1}^K w_a \exp \{ \ell_i(\mu_a) \} \right]. \quad (5.4)$$

EM initializes equal component weights except when refitting a replacement proposal, where the current weights are retained. The mean update is the responsibility-weighted average of  $q_i$ . The weight update exactly maximizes the EM auxiliary objective on  $\{w_a \geq 10^{-8}, \sum_a w_a = 1\}$ . No additional floor on means is needed because the observations are positive. Final labels maximize  $\log w_a + \ell_i(\mu_a)$ .

The main deterministic initialization, denoted *global gain*, selects its first seed using the proposed top-subset rule. Specifically, for each node candidate  $c$ , it sums its largest

$$m_0 = \min \{ \lfloor 2n/(3K) \rfloor, n - K \} \quad (5.5)$$

scores over other nodes, excluding  $c$ , and selects the candidate with the largest sum. Subsequent seeds maximize the all-node marginal gain

$$\Delta(c) = \sum_i \max \left\{ \ell_i(q_c) - \max_{a \in S} \ell_i(q_a), 0 \right\}, \quad (5.6)$$

over unselected candidates. All observations remain available. Seed profiles are frozen during these  $K$  selections; one soft EM fit follows after all  $K$  seeds have been selected. This is the archived `global_gain_no_update` arm.

We compare it with the following alternatives. *Top-subset peeling* uses the same first-seed rule and repeatedly removes the selected candidate and its best-scoring subset. When  $M$  candidates and  $r$  components remain, the subset contains  $\min \{ \lfloor 2M/(3K) \rfloor, M - r \}$  other nodes. After  $K$

selections, all  $n$  observations, including unselected nodes, enter EM. *Peeling plus replacement* fits this initialization, deletes each component temporarily, proposes the globally most useful node against the remaining  $K - 1$  means, and refits all nodes. All  $K$  proposals use the same original fitted endpoint; the best proposal is accepted only if its final  $\mathcal{L}$  exceeds the current best by more than  $10^{-5}$ . This is one replacement pass, not repeated optimization.

*Growing global gain* starts from the global profile mean and runs EM after each added component. It differs from the main initializer even though it happens to produce the same recovery errors here. *Residual++* samples the first node uniformly and later nodes proportionally to

$$d_i = \min_{a \in S} \sum_j [q_{ij} \log(q_{ij}/\mu_{aj}) - q_{ij} + \mu_{aj}]. \quad (5.7)$$

This deviance is not squared again. Its greedy variant samples  $2 + \lfloor \log K \rfloor$  candidate trials per addition and chooses the trial that minimizes total residual deviance. We report both one start and ten starts; the ten-start fit is selected solely by final  $\mathcal{L}$ . An additional control selects one set of Euclidean  $k$ -means++ seeds in  $X = \sqrt{n}U\Lambda$  and then runs the same Poisson-profile EM. The main Euclidean baseline runs  $k$ -means in  $X$  with ten starts, at most 300 Lloyd iterations per start, and tolerance  $10^{-5}$ .

All profile EM fits use at most 300 updates and tolerance  $10^{-6}$ . The stopping rule requires unchanged hard labels, maximum relative mean or absolute weight change at most  $10^{-3}$ , and objective increment at most  $10^{-6}(1 + |\mathcal{L} - G|)$  in absolute value, where  $G = \sum_{ij} \log \Gamma(q_{ij} + 1)$  is a cached observation-only constant. This constant sets the stopping scale and does not affect fitting or selection. Randomized seeding uses fixed NumPy seed sequences indexed by graph seed, method, and restart. Reported error rates minimize label disagreement over all permutations of the  $K$  labels. Averages weight graphs equally; “exact” counts graphs with no mislabeled node. True labels are accessed only after all fitting and likelihood-based selections finish.

### 5.3 Recovery results

Table 1 gives the paired results in the design and independent batches. Global-gain initialization, peeling plus replacement, growing global gain, and ten-start greedy residual++ have the same per-graph error counts in both batches. Across all 240 graphs, the main deterministic method has average error 0.0946%, compared with 0.3194% for the ten-start Euclidean baseline, and exactly recovers 163 graphs compared with 139. These are finite-sample recovery comparisons with unequal optimization budgets, rather than equal-runtime comparisons.

Table 2 separates model families. The largest improvement over Euclidean decoding occurs with heterogeneous diagonal connectivities: the average error falls from 1.4175% to 0.1625%. The likelihood criterion also improves the unequal-size model. It is not uniformly better: the symmetric two-community baseline has one fewer mislabeled node in aggregate, and the hierarchical model has one additional exactly recovered graph under  $k$ -means. The examples therefore support likelihood adaptation when block profiles differ substantially, without claiming dominance on every graph.

The stronger signal coefficient  $\eta = 1.4$  gives 117 exactly recovered graphs out of 120 for global gain, versus 99 for  $k$ -means. At  $\eta = 0.8$ , the corresponding counts are 46 and 40. This behavior is consistent with the importance of the logarithmic-density recovery threshold, but the experiment does not establish that threshold or the leading error exponent.

Table 1: Average mislabeled fraction in percent; exact recovery counts are out of 120 graphs per batch. All profile arms use the same EM score.

| Method                             | Design batch  |       | Independent batch |       |
|------------------------------------|---------------|-------|-------------------|-------|
|                                    | Error (%)     | Exact | Error (%)         | Exact |
| Top-subset peeling + EM            | 0.3917        | 86    | 0.3725            | 77    |
| Peeling + replacement              | <b>0.0967</b> | 86    | <b>0.0925</b>     | 77    |
| Global gain + EM                   | <b>0.0967</b> | 86    | <b>0.0925</b>     | 77    |
| Growing global gain + EM           | 0.0967        | 86    | 0.0925            | 77    |
| Growing sampled gain + EM          | 0.3900        | 86    | 0.0925            | 77    |
| Residual++, one start + EM         | 4.0129        | 80    | 3.1675            | 72    |
| Greedy residual++, one start + EM  | 1.1192        | 85    | 2.3438            | 74    |
| Greedy residual++, ten starts + EM | 0.0967        | 86    | 0.0925            | 77    |
| Euclidean++, one start + EM        | 3.9033        | 78    | 6.4204            | 71    |
| $X$ $k$ -means, ten starts         | 0.3454        | 73    | 0.2933            | 66    |

Table 2: Both batches combined, 40 graphs per model. “Global gain” refers to the top-subset first seed followed by frozen-profile all-node gains and one final EM fit.

| Model                          | Global gain + EM |       | $X$ $k$ -means |       |
|--------------------------------|------------------|-------|----------------|-------|
|                                | Error (%)        | Exact | Error (%)      | Exact |
| Symmetric two-community        | 0.0150           | 33    | 0.0125         | 34    |
| Equal-size three-community     | 0.0175           | 32    | 0.0188         | 31    |
| Heterogeneous three-community  | 0.1625           | 18    | 1.4175         | 1     |
| Unequal-size three-community   | 0.0338           | 33    | 0.1213         | 25    |
| Hierarchical four-community    | 0.0275           | 28    | 0.0288         | 29    |
| Disassortative three-community | 0.3113           | 19    | 0.3175         | 19    |

## 5.4 A failure of subset scores and its repair

One design graph with  $n = 2000$ ,  $K = 3$ , and  $\eta = 0.8$  illustrates the initialization mechanism. Its normalized connectivity matrix is approximately

$$C = \begin{pmatrix} 7.8541 & 0.7854 & 1.5708 \\ 0.7854 & 8.6395 & 0.7854 \\ 1.5708 & 0.7854 & 7.8541 \end{pmatrix}, \quad (5.8)$$

with group sizes (666, 666, 668). Communities 1 and 3 are the closest pair: their empirical  $X$ -centroid distance is 40.22, compared with 47.36 and 47.82 for the other pairs. They are nevertheless well separated in the retained spectrum. Using their true profile means as a diagnostic classifier assigns every node correctly on its first assignment.

Top-subset peeling selects nodes 858, 610, 435 (zero-based IDs), whose true communities are 2, 3, 2. The profile of node 610 lies near the mixture of communities 1 and 3. Its Poisson divergence to their merged mean is 0.3826, compared with 5.2433 to its own community’s mean. Its selected subset has 150 nodes from community 1 and 196 from community 3. Thus a candidate can win a subset likelihood comparison by representing a mixture of communities. The resulting fit merges communities 1 and 3 and splits community 2, making 709 errors (35.45%). Tightening the tolerance to  $10^{-10}$  and permitting 3000 updates does not solve this structural problem: the fit stops after 517 updates with 800 errors.

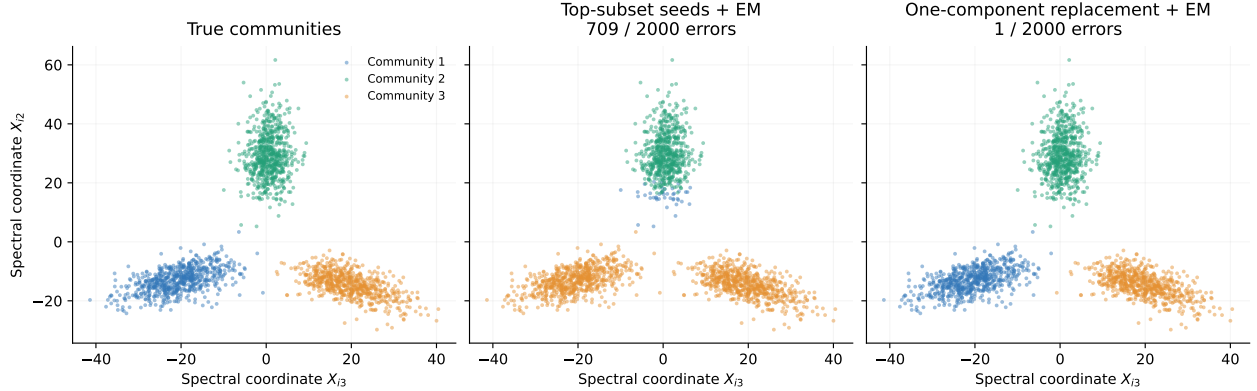


Figure 1: The saved design failure, displayed in two coordinates of  $X = \sqrt{n}U\Lambda$ . Left: true communities, used only for evaluation. Middle: top-subset peeling followed by EM merges the two lower communities and splits the upper community. Right: an accepted one-component replacement followed by EM separates the groups and leaves one mislabeled node. Predicted labels are aligned to truth for display only.

Global gain keeps the first two seeds but changes the third to node 220, which contributes to the uncovered region; after EM, only one node is mislabeled. A one-component replacement at the original fitted endpoint selects the same node and likewise reduces errors to one, while increasing the working log likelihood by approximately 7148.48. No true labels enter either choice. The independent batch contains a second analogous peeling failure with 675 errors, reduced to three by replacement. Relative to peeling, each effective deterministic variant improves exactly one graph in each batch, leaves the other 119 error counts unchanged, and worsens none. The number of exact recoveries does not change because the repaired graphs retain one and three errors, respectively.

## 5.5 Numerical verification and computational scope

All returned profile fits satisfy the declared stopping rule in the 240-graph comparison. The 120 original peeling errors are reproduced exactly. The archive contains 6240 EM traces, including replacement proposals, growing stages, and restart fits, with 27867 recorded trace points. The smallest observed objective increment is  $-1.17 \times 10^{-10}$ , within floating-point rounding; no increment below  $-10^{-7}$  occurs. All 240 selected ten-start solutions maximize the same final working likelihood within their restart set. Replacement accepts a likelihood improvement on 22 design graphs and 25 independent graphs, and never lowers the final objective; most of these small improvements do not change the classification error. Independent scalar checks reproduce deviance and global-gain calculations with maximum absolute errors  $7.11 \times 10^{-15}$  and  $1.78 \times 10^{-15}$ , respectively.

The present implementation explicitly reconstructs the  $n \times n$  profiles and caches all singleton-candidate scores. Reconstruction costs  $O(n^2K)$ , score precomputation costs  $O(n^3)$ , and storage costs  $O(n^2)$ . An EM update costs  $O(n^2K)$ . A replacement pass tries  $K$  additional fits, whereas ten-start residual seeding uses ten fits. The runs use one numerical library thread per worker and four worker processes. Python, NumPy, SciPy, and scikit-learn versions are 3.11.7, 2.1.3, 1.16.2, and 1.7.1. Stored timers cover all candidate arms and restarts together and exclude graph generation and eigenpair computation, so they do not support an arm-specific speed claim. The data sizes  $n \leq 2000$  establish feasibility for this implementation; candidate-pool approximations and blockwise computations require separate large-graph validation.

The evidence supports improved decoding of a fixed spectrum and a repair of redundant initialization under the declared working likelihood. It does not by itself prove that the practical EM endpoint has the oracle error exponent, that clipping preserves exact graph likelihood ratios, or that recovery is uniform over general SBMs. Oracle-LRT comparisons, a denser range of  $n$  for error-exponent estimation, and lower-density regularized experiments remain outside the archived numerical study.

## 6 Conclusion

The likelihood-relevant information in an SBM is organized by block-constant logarithmic contrasts. A full-rank population spectral representation contains these contrasts, and the projection identity explains how their adjacency-row statistics can be preserved by spectral compression. This leads to a concrete spectral-only decoder: choose one profile on a high-likelihood subset, add profiles by joint likelihood gain, and alternate mixture fitting and classification. A complementary parameter-replacement step addresses a failure in which two communities are merged while a third is split.

The empirical results support the usefulness of changing the decoding objective. The proven results characterize exact contrast preservation, the leading sparse likelihood approximation, noiseless seed coverage, greedy gain, objective convergence under exact updates, and conditional transfer of the oracle error exponent. Together they separate three issues that otherwise look similar: optimizing a working likelihood, retaining likelihood information through compression, and attaining a statistical recovery rate.

The remaining central question is whether the empirical clipped-profile procedure satisfies the score-transfer conditions with exceptional-event probabilities at the required exponential scale. Its successful finite-grid experiments do not settle that question. Low-density calibration, initialization stability, parameter estimation, and a scalable implementation provide further work. A proof of those conditions would turn the conditional  $e^{-(1+o(1))I_n}$  recovery statement into an end-to-end theorem for the executable algorithm.

## A Proofs of spectral preservation and statistical results

### A.1 Exact preservation and the population contrast space

*Proof of Proposition 3.1.* Since  $AU = U\Lambda$ , multiplication by  $U^\top$  gives  $AQ = AUU^\top = U\Lambda U^\top = \tilde{P}$ . Therefore  $\tilde{P}_i w - A_i w = A_i(Q - I)w$ . If  $w \in \text{col}(U)$ , then  $Qw = w$  and the difference vanishes.

Let  $N = Z^\top Z = \text{diag}(n_1, \dots, n_K)$ , which is nonsingular since all communities are nonempty. The columns of  $ZN^{-1/2}$  are orthonormal, and

$$P^\circ = (ZN^{-1/2})\{\rho_n N^{1/2} B N^{1/2}\}(ZN^{-1/2})^\top.$$

Nonsingularity of  $B$  makes the central matrix nonsingular. Hence  $P^\circ$  has rank  $K$  and column space  $\text{col}(Z)$ . For each contrast, define  $\xi_c^{ab} = \log\{B_{ac}(1 - \rho_n B_{bc})/[B_{bc}(1 - \rho_n B_{ac})]\}$ . Then  $w^{ab} = Z\xi^{ab}$ , which proves membership in that space. Finally, subtracting the diagonal of  $P^\circ$  gives the actual no-self-loop mean, so no rank assertion about that corrected matrix is needed.  $\square$

Full rank cannot be replaced by distinct block rows without a further contrast-span condition. For example, let  $B = ss^\top$  for a positive, nonconstant vector  $s$ . Its signal space is spanned by  $Zs$ ,

whereas the Poisson contrast between communities  $a, b$  is  $\log(s_a/s_b)\mathbf{1}$ . When  $s_a \neq s_b$ , this constant vector does not belong to  $\text{span}(Zs)$ . A rank-one signal projection does not preserve that contrast by the preceding identity. This example only concerns projection onto the population rank-one space; it is not an impossibility claim for every statistic computed from one empirical eigenvector.

## A.2 The sparse likelihood correction

*Proof of Lemma 3.2.* Write  $p_j = \rho_n B_{a,z_j}$  and  $q_j = \rho_n B_{b,z_j}$  for  $j \neq i$ . The exact binary Bernoulli log-likelihood ratio is

$$L_i^{ab} = \sum_{j \neq i} A_{ij} \log \frac{p_j}{q_j} + \sum_{j \neq i} A_{ij} \log \frac{1 - q_j}{1 - p_j} + \sum_{j \neq i} \log \frac{1 - p_j}{1 - q_j}.$$

For  $0 \leq x \leq 1/2$ , the power series for  $\log(1 - x)$  implies  $|\log(1 - x) + x| \leq Cx^2$ . Since  $p_j, q_j \leq b_+ \rho_n$ , it also implies  $|\log((1 - q_j)/(1 - p_j))| \leq C\rho_n$  and

$$\left| \log \frac{1 - p_j}{1 - q_j} + (p_j - q_j) \right| \leq C\rho_n^2.$$

The first term in the exact LLR is the Poisson linear term, and the linear approximation to its final term is the Poisson offset. Summing the two displayed remainders proves (3.6).

Under any true community,  $\deg(i)$  is a sum of independent Bernoulli variables with mean at most  $b_+ d_n$ . For  $D > b_+$ , exponential Markov inequality with  $t = \log(D/b_+)$  and the bound  $\log \mathbb{E} e^{t \deg(i)} \leq b_+ d_n (e^t - 1)$  give

$$\Pr\{\deg(i) > D d_n\} \leq \exp\{-d_n[D \log(D/b_+) - D + b_+]\}.$$

The bracket tends to infinity with  $D$ , so  $D_H$  can be chosen for any  $H$ . On the complementary event the correction is at most  $C(D_H + 1)d_n \rho_n = o(d_n)$ . If  $d_n = \log n$ , this becomes  $O(\log^2 n/n)$ .  $\square$

## A.3 Chernoff information and the oracle exponent

*Proof of Lemma 3.3.* Let  $f_a, f_b$  denote the product Bernoulli mass functions of the incident-edge row in the two hypotheses. Define

$$\begin{aligned} C_{i,ab}(t) &= -\log \sum_x f_a(x)^{1-t} f_b(x)^t \\ &= -\sum_{j \neq i} \log \{p_j^{1-t} q_j^t + (1 - p_j)^{1-t} (1 - q_j)^t\}, \\ C_{i,ab} &= \max_{0 \leq t \leq 1} C_{i,ab}(t). \end{aligned}$$

Uniformly for  $t \in [0, 1]$  and  $u, v \in [b_-, b_+]$ ,

$$(\rho_n u)^{1-t} (\rho_n v)^t + (1 - \rho_n u)^{1-t} (1 - \rho_n v)^t = 1 - \rho_n \{(1 - t)u + tv - u^{1-t} v^t\} + O(\rho_n^2).$$

For example, the nonedge product expansion follows by writing it as  $\exp\{(1 - t)\log(1 - \rho_n u) + t\log(1 - \rho_n v)\}$ ; all first and second derivatives involved are uniformly bounded on the specified compact set. Expanding  $-\log(1 - x)$  and summing gives

$$C_{i,ab} = d_n \max_{0 \leq t \leq 1} \sum_c \pi_{c,n} \{(1 - t)B_{ac} + tB_{bc} - B_{ac}^{1-t} B_{bc}^t\} + O(n\rho_n^2 + \rho_n), \quad (\text{A.1})$$

where  $\pi_{c,n} = n_c/n$ . The  $O(\rho_n)$  term accounts for the excluded node. Uniform convergence of the expression on  $[0, 1]$ ,  $\pi_{c,n} \rightarrow \pi_c$ , and  $\rho_n \rightarrow 0$  imply  $C_{i,ab} = d_n J_{ab} + o(d_n)$ . Distinct block rows and positive community proportions imply  $J_{ab} > 0$ .

The equal-prior Bayes risk is  $R_{i,ab}^{\text{or}} = \frac{1}{2} \sum_x \min\{f_a(x), f_b(x)\}$ . Since  $\min\{u, v\} \leq u^{1-t} v^t$ ,

$$R_{i,ab}^{\text{or}} \leq \frac{1}{2} e^{-C_{i,ab}}. \quad (\text{A.2})$$

For completeness, a logarithmic lower bound follows by exponential tilting, without invoking any refined rate formula. Let  $t_*$  maximize  $C_{i,ab}(t)$  and set

$$f_*(x) = e^{C_{i,ab}} f_a(x)^{1-t_*} f_b(x)^{t_*}, \quad L(x) = \log \frac{f_a(x)}{f_b(x)}.$$

All masses are positive, the two distributions differ, and the endpoint derivatives have opposite signs. Thus  $t_* \in (0, 1)$ , and differentiating the normalization gives  $\mathbb{E}_* L = 0$ . The tilted incident edges remain independent, with probabilities

$$p_{*,j} = \frac{p_j^{1-t_*} q_j^{t_*}}{p_j^{1-t_*} q_j^{t_*} + (1-p_j)^{1-t_*} (1-q_j)^{t_*}} \leq C \rho_n.$$

The coefficient of each edge in  $L$  is bounded by a constant depending only on  $b_-, b_+$ ; hence  $\text{Var}_*(L) \leq C d_n$ . Taking  $h_n = d_n^{2/3}$ , Chebyshev's inequality gives  $\Pr_*(|L| \leq h_n) = 1 - o(1)$ . The change of measure yields

$$\begin{aligned} R_{i,ab}^{\text{or}} &= \frac{1}{2} e^{-C_{i,ab}} \mathbb{E}_* \min\{e^{t_* L}, e^{-(1-t_*)L}\} \\ &\geq \frac{1}{2} e^{-C_{i,ab} - h_n} \Pr_*(|L| \leq h_n). \end{aligned}$$

Combining this with (A.2) proves  $\log R_{i,ab}^{\text{or}} = -C_{i,ab} + o(d_n)$ , and hence (3.9). For priors  $\nu, 1 - \nu$  bounded away from zero,  $\min\{\nu f_a, (1 - \nu) f_b\}$  is bounded above and below by constant multiples of  $\min\{f_a, f_b\}$ , so its leading exponent is unchanged.  $\square$

#### A.4 Transfer of the leading exponent

*Proof of Theorem 3.4.* Suppose the true label of node  $i$  is  $a$ . If  $G_i$  occurs and the classifier prefers  $b \neq a$ , then  $\hat{\ell}_i(a) - \hat{\ell}_i(b) \leq 0$ , and consequently  $L_i^{ab} \leq \tau_n$ . For every  $t \in [0, 1]$ ,

$$\Pr_a\{L_i^{ab} \leq \tau_n\} \leq e^{t\tau_n} \mathbb{E}_a e^{-tL_i^{ab}} = \exp\{t\tau_n - C_{i,ab}(t)\}.$$

Selecting a Chernoff maximizer and using  $t \leq 1$  gives

$$\Pr\{\hat{z}_i \neq z_i\} \leq \delta_n + \sum_{b \neq a} \exp\{\tau_n - C_{i,ab}\} \leq \delta_n + (K - 1) \exp\{-I_n + o(d_n)\}.$$

Since  $J_* > 0$  is fixed,  $o(d_n) = o(I_n)$ , and (3.12) makes the first term no larger in leading exponential order. Averaging over nodes and using that the permutation-invariant error is no larger than the error under any one common alignment proves (3.13).

When  $d_n = \log n$ , the expected number of errors under that alignment is at most  $n^{1-J_*+o(1)}$ . If  $J_* > 1$ , Markov's inequality gives vanishing probability of even one error, which proves the exact recovery assertion. No conclusion at the boundary  $J_* = 1$  follows from this argument.  $\square$

The matching binary lower bound mentioned after the theorem has a direct decision-theoretic justification. Conditional on all other labels, the rest of the graph is independent of the incident-edge row and has the same distribution under  $a$  and  $b$ . Revealing the other labels and true parameters cannot increase the Bayes risk. A function of the graph's spectral summary is also a function of the full graph; its risk in that binary experiment is therefore at least  $R_{i,ab}^{\text{or}}$ . This argument does not, by itself, assert a global minimax lower bound over an unspecified parameter space.

## A.5 Row-independent projection errors

*Proof of Theorem 3.5.* Condition on the data generating  $Q_i$  and on  $H_i$ . Work in the  $n-1$  coordinates other than  $i$ , and write  $Q_* = Q_{*,i}$ ,  $Q = Q_i$ ,  $w = w_{-i}^{ab}$ ,  $p_j = \mathbb{E}A_{ij}$ , and  $e = (Q - I)w$ . The fixed block assumptions imply

$$w \in \text{col}(Q_*), \quad p \in \text{col}(Q_*), \quad \|w\|_2 \leq C\sqrt{n}, \quad \|w\|_\infty \leq C, \quad \|p\|_2 \leq C\rho_n\sqrt{n}.$$

The constants are uniform over the finitely many contrasts and true labels. For orthogonal projectors,

$$\|Q_*(I - Q)Q_*\|_{\text{op}} = \|(I - Q)Q_*\|_{\text{op}}^2 \leq \eta_n^2,$$

because the left matrix is  $[(I - Q)Q_*]^\top [(I - Q)Q_*]$ . It follows that the mean projection error has the quadratic bound

$$|p^\top e| \leq \|p\|_2 \|w\|_2 \eta_n^2 \leq Cd_n \eta_n^2. \quad (\text{A.3})$$

Also,  $\|e\|_2 \leq C\sqrt{n}\eta_n$ . By Cauchy-Schwarz and projector leverage,

$$|(Qw)_j| \leq \|Qe_j\|_2 \|w\|_2 = \sqrt{Q_{jj}} \|w\|_2 \leq C\sqrt{L},$$

so  $\|e\|_\infty \leq M$  for a fixed positive constant  $M$ .

Conditional independence makes  $X_j = (A_{ij} - p_j)e_j$  independent and centered. They satisfy  $|X_j| \leq M$  and

$$V = \sum_j \mathbb{E}X_j^2 \leq b_+ \rho_n \|e\|_2^2 \leq Cd_n \eta_n^2. \quad (\text{A.4})$$

Here a Bennett bound is stronger than the simpler Bernstein bound at the required scale. We include its short derivation. For a centered  $X$  with  $|X| \leq M$  and  $\lambda \geq 0$ , Taylor expansion and  $|X|^k \leq M^{k-2}X^2$  imply

$$\mathbb{E}e^{\lambda X} \leq 1 + \frac{\mathbb{E}X^2}{M^2}(e^{\lambda M} - 1 - \lambda M).$$

After multiplying moment-generating functions and using  $\log(1+x) \leq x$ , exponential Markov inequality optimized in  $\lambda$  gives

$$\Pr\{|\sum_j X_j| > t \mid Q\} \leq 2 \exp\left\{-\frac{V}{M^2} h\left(\frac{Mt}{V}\right)\right\}, \quad h(u) = (1+u)\log(1+u) - u, \quad (\text{A.5})$$

with the evident interpretation when  $V = 0$ .

To choose a uniform vanishing error scale, put  $a_n = \max\{\eta_n^2, e^{-d_n}\}$ ,  $s_n = \log(1/a_n)$ , and  $\varepsilon_n = s_n^{-1/2}$ . Both  $a_n \rightarrow 0$  and  $s_n \rightarrow \infty$ . Take  $t = d_n \varepsilon_n$ . The bound (A.5) remains valid with  $V$  replaced by the upper bound  $Cd_n a_n$ , since its exponent is nonincreasing in the variance argument. As  $u \rightarrow \infty$ ,  $h(u) \sim u \log u$ , so

$$\frac{Cd_n a_n}{M^2} h\left(\frac{M\varepsilon_n}{Ca_n}\right) = \frac{d_n \varepsilon_n}{M} \{s_n + \log \varepsilon_n + O(1)\}(1 + o(1)) \geq cd_n \sqrt{s_n}$$

for all sufficiently large  $n$ . Combining this with (A.3) gives, for  $\tau_n = d_n \varepsilon_n + C d_n \eta_n^2 = o(d_n)$ ,

$$\Pr\{|A_{i,-i}(Q - I)w| > \tau_n \mid Q, H_i\} \leq 2e^{-cd_n \sqrt{s_n}}.$$

Union bound over the finitely many contrasts and addition of  $\Pr(H_i^c)$  prove (3.17) with  $h_n = c\sqrt{s_n} \rightarrow \infty$ . The last claim is an application of Theorem 3.4.  $\square$

## B Proofs of initialization and optimization properties

### B.1 Frozen-candidate greedy gains

**Proposition B.1** (Greedy gain guarantee). *Let  $s_{ic}$  be fixed finite candidate scores and let  $S_0$  be a fixed nonempty seed set. For candidate sets  $T$ , define*

$$G(T) = \sum_i \max_{c \in S_0 \cup T} s_{ic} - \sum_i \max_{c \in S_0} s_{ic}.$$

*Then  $G$  is normalized, monotone, and submodular. If  $m \geq 1$  greedy steps select a set  $T_m$  and  $T^*$  maximizes  $G$  over sets of at most  $m$  candidates, then*

$$G(T_m) \geq \{1 - (1 - 1/m)^m\} G(T^*) \geq (1 - e^{-1}) G(T^*).$$

*Proof.* Normalization and monotonicity follow immediately from the definition. The marginal gain of candidate  $c$  is

$$G(T \cup \{c\}) - G(T) = \sum_i \left( s_{ic} - \max_{b \in S_0 \cup T} s_{ib} \right)_+.$$

If  $T \subseteq T'$ , each maximum in the right side increases, so the marginal gain decreases. This proves submodularity. For the greedy set  $T_t$  at time  $t$ , monotonicity and telescoping diminishing returns give

$$\begin{aligned} G(T^*) - G(T_t) &\leq G(T_t \cup T^*) - G(T_t) \\ &\leq \sum_{c \in T^* \setminus T_t} \{G(T_t \cup \{c\}) - G(T_t)\} \\ &\leq m \{G(T_{t+1}) - G(T_t)\}. \end{aligned}$$

Starting from  $G(T_0) = 0$  and iterating this inequality proves the claimed factor. The final inequality follows from  $(1 - 1/m)^m \leq e^{-1}$ , including the case  $m = 1$ .  $\square$

This is the standard cardinality-constrained submodular greedy guarantee [7]. It controls added explanation relative to the fixed initial seed, not the multiplicative approximation of a possibly negative total log score. It does not apply unchanged when intermediate EM steps alter the candidate scores.

### B.2 Noiseless profile selection

**Proposition B.2** (Population profile selection). *Suppose  $q_i = \theta_{z_i}$ , where the  $K$  positive profile vectors  $\theta_1, \dots, \theta_K$  are pairwise distinct and every class is nonempty. The candidate pool consists of these node profiles. From any first seed,  $K - 1$  subsequent global-gain greedy steps select all  $K$  different profiles. Maximum working-score classification using those profiles recovers  $z$  up to permutation.*

*Proof.* For  $x, \mu > 0$  coordinatewise, write  $\ell_x(\mu) = \sum_j x_j \log \mu_j - \sum_j \mu_j$ . Then

$$\ell_x(x) - \ell_x(\mu) = \sum_j \{x_j \log(x_j/\mu_j) - x_j + \mu_j\} = D_P(x\|\mu) \geq 0,$$

with equality precisely when  $x = \mu$ ; the inequality follows from  $\log u \leq u - 1$ . If a profile has already been selected, selecting a duplicate has zero gain. If profile  $\theta_b$  has not been selected, then each node in class  $b$  gains  $\min_{a \in S} D_P(\theta_b\|\theta_a) > 0$  when it is added. Therefore some unselected profile has strictly positive gain, and greedy cannot select a duplicate before all distinct profiles have been selected. The same strict divergence inequality makes each node's own profile its unique best score after selection.  $\square$

The proposition concerns an abstract noiseless profile model. It does not replace a stability analysis for random spectral profiles, and it does not erase the diagonal distinction in a no-self-loop SBM.

### B.3 Exact working EM and accepted replacements

**Proposition B.3** (Objective convergence of exact working EM). *For fixed strictly positive profiles  $q_1, \dots, q_n$ , let*

$$\mathcal{L}(\omega, \mu) = \sum_i \log \sum_{a=1}^K \omega_a \exp\{\ell_i(\mu_a)\}, \quad \ell_i(\mu_a) = \sum_j q_{ij} \log \mu_{aj} - \sum_j \mu_{aj}.$$

*Starting with positive parameters and positive mixture weights, the exact E-step and unconstrained exact M-step*

$$r_{ia} = \frac{\omega_a e^{\ell_i(\mu_a)}}{\sum_b \omega_b e^{\ell_i(\mu_b)}}, \quad N_a = \sum_i r_{ia},$$

$$\omega_a^{\text{new}} = N_a/n, \quad \mu_{aj}^{\text{new}} = \frac{\sum_i r_{ia} q_{ij}}{N_a}$$

*make  $\mathcal{L}$  nondecreasing. Its numerical values converge to a finite limit. The same monotonicity statement holds for an exact M-step maximizing over any stated constraint set containing the current parameters, and for replacement proposals accepted only when their final  $\mathcal{L}$  improves.*

*Proof.* For row-wise probability vectors  $r_i$ , define the variational objective

$$\mathcal{F}(r, \omega, \mu) = \sum_{i,a} r_{ia} \{\log \omega_a + \ell_i(\mu_a) - \log r_{ia}\}.$$

The log-sum inequality gives  $\mathcal{F}(r, \omega, \mu) \leq \mathcal{L}(\omega, \mu)$ , with equality for the E-step responsibilities. For fixed  $r$ , maximizing  $\sum_a N_a \log \omega_a$  over the simplex gives  $\omega_a = N_a/n$ . For each profile coordinate, maximizing  $(\sum_i r_{ia} q_{ij}) \log \mu_{aj} - N_a \mu_{aj}$  gives the displayed weighted mean. These are interior maxima because all inputs and responsibilities are positive. Thus the E-step makes the variational bound tight and the M-step raises it, proving monotonicity.

The saturated-score inequality used in the preceding proof implies  $\ell_i(\mu_a) \leq \ell_i(q_i)$  for every  $i, a$ . Since  $\sum_a \omega_a = 1$ ,

$$\log \sum_a \omega_a e^{\ell_i(\mu_a)} \leq \ell_i(q_i).$$

Consequently  $\mathcal{L} \leq \sum_i \ell_i(q_i) < \infty$ . A nondecreasing real sequence bounded above has a finite limit. For a constrained exact M-step the current parameters remain feasible, so the same variational argument applies. The replacement assertion follows directly from its acceptance rule.  $\square$

This proposition proves convergence of objective values. It proves neither convergence to the global maximum nor recovery of the true labels. The implementation uses an exact active-set maximizer on the floored simplex, so its weight constraint is covered by the constrained version of this proposition. Its finite stopping tolerance and floating-point accuracy are assessed separately in Section 5.

## C Implementation and reproducibility

### C.1 Experimental entry points and retained outputs

The numerical evidence is archived in the repository

<https://github.com/zhixin0825/spectral-lrt>.

The principal comparison is in `experiments/20261009_residual_likelihood_seeding/`. Its report, protocol, model-generation script, fitting code, scalar verification, and per-graph summaries define the numerical experiment. The preceding partial-peeling experiment and its diagnosis are in `experiments/20261009_partial_likelihood_peeling/`. Algorithm 1 corresponds to the recorded `global_gain_no_update` branch. The API option `method='global_gain'` instead denotes the growing-EM branch, which starts at a global profile mean. These are different algorithms even when their final error counts agree.

The manuscript’s two CSV summaries retain batch-specific, combined, and model-specific results. The diagnostic figure uses saved coordinates and fitted labels rather than a new simulation. Full spectral arrays and objective traces are stored in the previously archived numerical records; the source reports describe their restoration. Reconstructing the profiles from  $(U, \Lambda)$  and fitting the working score does not need the original adjacency matrix.

### C.2 Floored-simplex weight updates

For a fixed floor  $f \in [0, 1/K)$ , the weight M-step solves

$$\max_{\omega_a \geq f, \sum_a \omega_a = 1} \sum_a N_a \log \omega_a. \quad (\text{C.1})$$

Its active-set solution starts with every component free. If the free set is  $S$ , assign mass  $1 - f(K - |S|)$  proportionally to the  $N_a$ ,  $a \in S$ , and fix any weight below  $f$  at the floor. Repeat on the remaining free set. The KKT conditions give

$$\omega_a = \max\{f, N_a/\lambda\}, \quad \sum_a \omega_a = 1 \quad (\text{C.2})$$

for a suitable multiplier  $\lambda$ . The active-set procedure enforces this equation; it is not a naive clipping followed by renormalization. The tested function `_floored_weights` implements this exact constrained update. Since the E-step and mean M-step are unchanged, the variational proof applies on the floored-simplex parameter domain.

### C.3 An explicit error decomposition

For fitted components matched to labels  $a, b$ , put

$$\hat{w}_j^{ab} = \log(\mu_{aj}/\mu_{bj}), \quad \hat{c}^{ab} = -\sum_j (\mu_{aj} - \mu_{bj}) + \log(\omega_a/\omega_b). \quad (\text{C.3})$$

If  $\lambda^{ab}$  is the chosen oracle prior log ratio, the empirical comparison has the decomposition

$$\begin{aligned} \hat{T}_i^{ab} - (I_i^{ab} + \lambda^{ab}) &= A_i(\Pi - I)w^{ab} + (q_i - \tilde{P}_i)w^{ab} \\ &\quad + q_i(\hat{w}^{ab} - w^{ab}) + \hat{c}^{ab} - c_i^{ab} - \lambda^{ab}. \end{aligned} \quad (\text{C.4})$$

The terms correspond to empirical projection, positivity clipping, estimated contrasts, and intercept calibration. The reconstructed diagonal is included in these terms; the oracle excludes it through  $A_{ii} = 0$  and the definition of  $c_i^{ab}$ . For Poisson rather than Bernoulli oracle contrasts, the sparse correction must also be included. Formula (C.4) is exact and gives concrete quantities for checking the transfer conditions. The present manuscript does not establish the required simultaneous bounds for the practical EM endpoint.

### C.4 Verification of the manuscript

The source comprises six numbered main sections, proof appendices, and a bibliography. A build runs PDFLaTeX, BibTeX, and two additional PDFLaTeX passes, checking unresolved citations and references. Build provenance records the source commit and the checksums of the source and compiled PDF. Compilation checks syntax and cross references; mathematical conclusions are exactly those stated with their hypotheses in the text.

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